INTELLIGENT CROP RECOMMENDATION SYSTEM INTEGRATING IOT, CLIMATE DATA AND MARKET PRICE FORECASTS FOR ECONOMIC VIABILITY IN SRI LANKAN PADDY CULTIVATION

Bandara R.M.A.S.

(IT21811104)

BSc (Hons) in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

INTELLIGENT CROP RECOMMENDATION SYSTEM INTEGRATING IOT, CLIMATE DATA AND MARKET PRICE FORECASTS FOR ECONOMIC VIABILITY IN SRI LANKAN PADDY CULTIVATION

Bandara R.M.A.S.

(IT21811104)

Dissertation submitted in partial fulfilment of the requirements for the Bachelor of Science (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

Project R26-IT-066

October 2026

DECLARATION

I declare that this is my own work and this dissertation does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any other university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

I also grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation, in whole or in part, in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Signature: …… Date: ……

The above candidate has carried out research for the bachelor's degree dissertation under my supervision.

Signature of the supervisor: …… Date: ……

(Ms. Chathurangika Kahandawarachchi)

Signature of the co-supervisor: …… Date: ……

(Ms. Buddhima Attanayake)

ABSTRACT

Paddy farmers in Sri Lanka choose varieties and planting plans under uncertain weather, uneven soil nutrients and unstable market prices, yet most digital advisory tools weigh only soil and climate. The individual component IT21811104 of Project R26-IT-066 addresses this gap with a market-aware paddy variety recommendation and price-forecasting component. The research follows a Design Science Research approach. The implemented artefact has three parts: the Smart Crop Guide screen of the AgriSmart Paddy Flutter application, a Flask prediction service, and a Firebase Realtime Database connection to the wider system. The screen accepts soil moisture, temperature, humidity, nitrogen, phosphorus and potassium values, the cultivation season, the planned harvest month and the current paddy market price. The service loads a Random Forest crop model and a market price model, predicts a price for each candidate variety, adds a fixed variety bonus and returns the top three varieties ranked by price.

Three tree-based classifiers (Random Forest, XGBoost and Decision Tree) were trained on a ten-column paddy dataset and compared on accuracy. XGBoost scored highest at about 0.44, followed by Decision Tree at about 0.39 and Random Forest at about 0.33. These values are far below the proposal target of 85% and are reported as a baseline, not as a success. The dissertation analyses the probable causes: a very small evaluation set, an undocumented classification target, a dataset in which roughly half of the yield values equal 2800, and the absence of price-model training data and metrics. The deployed crop model is Random Forest, the lowest-scoring of the three. The ranking depends on fixed variety bonuses, the suitability and net-revenue scoring are not implemented, and the harvest month and price inputs may not reach the backend. Latency and farmer usability results are not yet available. The artefact delivers a working prototype and a modular foundation for market-aware recommendation, but its recommendation quality is not yet sufficient for farmer-facing use.

Keywords – paddy variety recommendation, market price forecasting, Random Forest, XGBoost, Decision Tree, IoT, decision support, Sri Lankan paddy

ACKNOWLEDGEMENT

I sincerely thank my supervisor, Ms. Chathurangika Kahandawarachchi, and my co-supervisor, Ms. Buddhima Attanayake, for their guidance, feedback and encouragement throughout this research.

I acknowledge my R26-IT-066 group members, whose IoT field-monitoring, disease-analysis and harvest-yield components connect with my crop recommendation and market component. I also thank the academic staff of the Department of Information Technology at SLIIT, and the institutions, including the Department of Census and Statistics and the Department of Agriculture, whose publications informed this work.

Finally, I thank my family and friends for their constant support.

TABLE OF CONTENTS

DECLARATION ..... **ii**
ABSTRACT ..... **iii**
ACKNOWLEDGEMENT ..... **iv**
LIST OF FIGURES ..... **vii**
LIST OF TABLES ..... **viii**
LIST OF ABBREVIATIONS ..... **ix**
1. INTRODUCTION **1**
1.1 Background and Context 1
1.2 Literature Review 1
1.3 Comparison of Similar Work 2
1.4 Research Gap 3
1.5 Research Problem 3
1.6 Research Objectives 4
2. METHODOLOGY **5**
2.1 Research Approach 5
2.2 Overview of the Proposed System 5
2.3 Data Inputs and Sources 6
2.4 Data Preprocessing and Feature Preparation 7
2.5 Recommendation Models 7
2.6 Market Forecast and Scenario Design 8
2.7 Software Architecture and Implementation 8
2.8 Functional and Non-Functional Requirements 9
2.9 Testing Strategy 10
2.10 Commercialization Aspects 10
2.11 Methodological Limitations 10
2.12 Chapter Summary 10
3. RESULTS AND DISCUSSION **11**
3.1 Implemented Artifact Results 11
3.2 Dataset Analysis 14
3.3 Model Comparison 15
3.4 Market Forecast and Scenario Ranking 16
3.5 Integration with the Wider System 16
3.6 Functional Test Design 17
3.7 Non-Functional Evaluation 17
3.8 Research Findings 18
3.9 Comparison with Existing Approaches 18
3.10 Discussion of Research Objectives 18
3.11 Limitations 19
3.12 Practical Implications 19
3.13 Commercialization Findings 19
3.14 Chapter Summary 19

4. CONCLUSION.....20
4.1 Summary of the Research.....20
4.2 Achievement of the Aim.....20
4.3 Individual Contribution.....20
4.4 Key Technical Contribution.....20
4.5 Limitations and Required Corrections.....21
4.6 Future Work.....21
4.7 Final Conclusion.....21
REFERENCES.....22
GLOSSARY.....24
APPENDICES.....25
APPENDIX A – EVIDENCE INDEX.....25
APPENDIX B – SOURCE-CODE EVIDENCE: MODEL LOADING AND SERVICE SETUP.....25
APPENDIX C – SOURCE-CODE EVIDENCE: INPUT CONSTRUCTION.....26
APPENDIX D – SOURCE-CODE EVIDENCE: PRICE PREDICTION AND RANKING 27

LIST OF FIGURES

Figure 2.1: Component architecture: inputs to screen
Figure 3.1: Flutter launch on Edge in debug mode (device selection)
Figure 3.2: Flutter debug service and application start
Figure 3.3: Smart Crop Guide screen before a recommendation is generated
Figure 3.4: Smart Crop Guide screen showing the generated variety recommendation
Figure 3.5: Paddy dataset (first 25 records)
Figure 3.6: Model accuracy comparison (Random Forest, XGBoost, Decision Tree)
Figure B.1: Model loading and route definitions in the crop recommendation backend
Figure C.1: Request parsing and crop feature construction
Figure D.1: Price prediction, variety bonus and ranking logic

LIST OF TABLES

Table 1.1: Comparison of related research and agricultural systems
Table 1.2: Research gap analysis
Table 1.3: Specific objectives and measurable success criteria
Table 2.1: Mapping of research phases to objectives
Table 2.2: Input features and their roles
Table 2.3: Scenario logic: planned design and current implementation
Table 2.4: Functional requirements
Table 2.5: Non-functional requirements
Table 3.1: Classifier accuracy (read from Figure 3.6)
Table 3.2: Functional and integration test cases
Table 3.3: Evidence-based evaluation summary
Table 3.4: Objective achievement matrix
Table A.1: Evidence index

LIST OF ABBREVIATIONS

Abbreviation	Full form
API	Application Programming Interface
DCS	Department of Census and Statistics
DOA	Department of Agriculture, Sri Lanka
DSR	Design Science Research
DSS	Decision Support System
DT	Decision Tree
IoT	Internet of Things
IRRI	International Rice Research Institute
ML	Machine Learning
NPK	Nitrogen, Phosphorus and Potassium
REST	Representational State Transfer
RF	Random Forest
SES	Standard Evaluation System
UI	User Interface
WSI	Weighted Suitability Index
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION

1.1 Background and Context

Paddy is central to Sri Lankan food security and rural livelihoods. National paddy statistics are reported for the Maha and Yala seasons, and output depends on weather, water, nutrient management, crop health and variety choice [1].

Most farmers still choose what to plant from experience and static guidelines. These choices rarely reflect current field conditions or the price the crop will fetch at harvest. A variety that suits the soil can still lose money if prices fall or supply is high. Smart-farming research shows that sensing alone is not enough: data from many sources must be converted into actionable decisions [2], [3].

Project R26-IT-066 is a four-module Smart Agriculture Decision Support System (DSS). IT22178718 provides IoT field monitoring and fertilization guidance. IT21811104, the subject of this dissertation, provides crop recommendation and market analysis. IT22261632 provides disease detection and severity analysis. IT22266446 provides disease-aware harvest yield estimation. This dissertation covers only the IT21811104 contribution: using field and climate conditions together with market price information to recommend paddy varieties and forecast market outcomes.

1.2 Literature Review

Machine learning for crop recommendation. Liakos et al. reviewed machine learning in agriculture and found that yield prediction and crop management are mature application areas, but that usefulness depends on data quality and representativeness [4]. Abdurrahman and Miyim applied Random Forest to rice yield prediction in Nigeria and reported accuracy above 85% [5]. Their data come from a different climate, variety set and season structure, so the result supports the algorithm choice but not direct reuse for Sri Lankan varieties such as BG 300 and AT 362.

Tree-based ensemble models. Random Forest builds many randomized decision trees and averages them, which makes it robust to noise and non-linear relationships [6]. XGBoost builds trees sequentially with a regularized objective and is efficient on tabular data [7]. A single Decision Tree is easy to interpret but unstable on small data [8]. These three models are reasonable baselines for a tabular recommendation task, and they are the models compared in this research.

IoT-enabled field monitoring. Friha et al. surveyed IoT for smart agriculture and stressed sensing, communication and processing as parts of one pipeline [2]. Wang et al. showed that IoT sensor data can feed machine learning for rice disease detection [9]. In this project, the IoT

module supplies soil moisture, temperature, humidity and NPK values, which this component consumes as recommendation inputs.

Growth stage, disease and climate effects. Gumma et al. identified growth stage as an important variable for paddy outcomes [10]. Peng et al. quantified rice-blast yield losses under climate scenarios [11]. Chen et al. and Mandal et al. assessed rice-blast severity from images and spectral data [12], [13]. These studies justify using environmental and crop-stage variables, but none links them to a market-aware variety choice.

Sri Lankan platforms and institutional data. eKeth, GeoGoviya and Krushi Advisor show that digital advisory services are feasible in Sri Lanka [14], [15], [16]. They rely mainly on static data or generalized averages. The Department of Agriculture provides variety profiles and fertilizer guidelines [17], the Department of Census and Statistics publishes district-level paddy statistics [1], and the IRRI Standard Evaluation System gives reference ranges for rice assessment [18].

Price forecasting. Agricultural price forecasting normally relies on regression or time-series models trained on historical market prices. The proposal names the Hector Kobbekaduwa Agrarian Research Institute and government agrarian databases as price sources. This dissertation does not review price-forecasting studies in depth. The market component is therefore judged by its implementation and the evidence in Chapter 3, and a dedicated price-forecasting literature review is listed as future work.

1.3 Comparison of Similar Work

Table 1.1 compares the approaches most relevant to this component and shows what each leaves uncovered.

Table 1.1: Comparison of related research and agricultural systems

Related work or system	Main contribution	Scope not covered
DOA guidelines [17]	Variety profiles and fertilizer recommendations	Static documents; no live sensing or ML ranking
DCS paddy statistics [1]	District- and season-level production data	Retrospective; no field-level or price signal
IRRI SES [18]	Standard agronomic reference ranges	Not calibrated to BG or AT varieties; not an application
eKeth [14], GeoGoviya [15], Krushi Advisor [16]	Digital advisory for Sri Lankan farmers	No IoT input, price forecasting or economic scoring documented
Rice yield prediction, Nigeria [5]	Random Forest accuracy above 85%	Non-Sri Lankan data; no market component
IoT-ML disease detection [9]	Sensor pipeline for rice disease	Disease only; no recommendation or economics

Related work or system	Main contribution	Scope not covered
IT21811104 (this research)	Variety recommendation from field, season and price inputs with a Flutter interface	Accuracy below target; price model has no reported metrics; suitability and net-revenue scoring not implemented

1.4 Research Gap

Across the reviewed work, agronomic suitability and market viability are treated separately. The gap is the lack of an integrated, localized component that combines current field data, season and market price information into one recommendation a farmer can read.

Table 1.2: Research gap analysis

Gap	Existing limitation	Contribution of this research
G1 Market awareness	Recommendations ignore expected price and supply	Takes current market price and harvest month as interface inputs and presents a market forecast
G2 Live field data	Advisory tools use static or manually entered data	Accepts soil and climate values that the shared platform can supply from the IoT module
G3 Localization	Models trained outside Sri Lanka	Uses a Sri Lankan paddy dataset with BG 300 and AT 362 varieties and Maha and Yala seasons
G4 Presentation	Outputs are not ranked or scenario-based	Provides a farmer-facing screen that ranks three varieties with a forecast price

The contribution is deliberately narrow. This research does not claim that no agricultural software exists in Sri Lanka. It claims an integration of recommendation and market information within the wider DSS, and it reports how well that integration performs so far.

1.5 Research Problem

Crop choices in Sri Lanka rely on traditional knowledge and general guidelines rather than data. Existing recommenders use static soil and historical climate data and ignore real-time field conditions and economic factors. Farmers may therefore grow varieties that suit the land but return little money at harvest.

The main research problem is how to convert field conditions, season, harvest timing and market price information into an understandable paddy variety recommendation and market forecast for Sri Lankan smallholders. The problem includes a second question: whether the available data and standard tree-based models can reach a reliability that justifies farmer-facing use. This dissertation answers both by building the component and evaluating it on recorded experimental evidence.

1.6 Research Objectives

1.6.1 Main Objective

To develop and evaluate a market-aware, IoT-integrated paddy variety recommendation and price-forecasting component that uses soil, climate, season, harvest-month and market-price inputs to support planting decisions for Sri Lankan paddy farmers within the R26-IT-066 DSS.

1.6.2 Specific Objectives and Measurable Success Criteria

Table 1.3: Specific objectives and measurable success criteria

ID	Specific objective	Measurable success criterion (proposal)
SO1	Build a machine learning suitability classifier that recommends paddy varieties from IoT soil inputs and climate data, trained on Sri Lankan BG 300 and AT 362 data	Classification accuracy of at least 85% on a held-out validation set across Maha and Yala
SO2	Build a market viability regression model that forecasts paddy price and supply-demand pressure from historical market data	R^2 of at least 0.80 on held-out price data covering at least three consecutive seasons
SO3	Combine SO1 and SO2 into a three-scenario ranked recommendation (High Profit, Balanced Risk, Agronomic Safe Choice) with revenue projections	All three scenarios returned within 3 seconds; ranking consistency of at least 90% across repeated runs
SO4	Deliver the output through an application integrated with the group platform	At least 80% task completion among 10 farmers; API response within 3 seconds
SO5	Evaluate the component against these technical and functional criteria using documented evidence	Reported metrics, test records and an objective-achievement analysis

The criteria describe the evidence required. They do not state that each objective has passed. Chapter 3 reports which criteria are met, partly met or still open.

1.6.3 Summary

The objectives cover a suitability classifier, a market price model, a scenario ranking engine, an application interface and an evaluation of all four. Together they define the individual contribution of IT21811104 inside the Smart Agriculture DSS.

2. METHODOLOGY

This chapter describes how the IT21811104 component was designed, implemented and evaluated. It separates what the proposal planned from what the implementation and experiment records show, so that Chapter 3 can report results without overstating them.

2.1 Research Approach

The research follows Design Science Research (DSR) because its main output is a working software artefact built to solve a practical decision problem [19]. DSR moves through problem identification, objective definition, design and development, demonstration and evaluation. Table 2.1 maps these phases to the objectives.

Table 2.1: Mapping of research phases to objectives

Research phase	Main activity	Objective
Problem identification	Analyse crop-choice and market-uncertainty problems of paddy farmers	Overall
Data preparation	Identify field, season and market variables; prepare the paddy dataset	SO1, SO2
Model development	Train and compare Random Forest, XGBoost and Decision Tree; load a market price model	SO1, SO2
Market and scenario design	Define price-based ranking and the planned suitability and net-revenue scoring	SO2, SO3
Application integration	Build the Smart Crop Guide screen and the Flask prediction service	SO4
Evaluation	Measure model quality, test the interface and report gaps	SO5

2.2 Overview of the Proposed System

The component is the crop-recommendation and market layer of the R26-IT-066 DSS. It receives soil and climate values, combines them with the cultivation season, harvest month and current market price entered on the interface, and returns ranked variety recommendations with a forecast price. The proposal also plans to share the current market price with the harvest-yield module (IT22266446). The IT22266446 dissertation does not yet use market price in its decision route, so this link is not yet implemented.

The component has four internal layers: data inputs, preprocessing and feature preparation, prediction models, and the farmer interface. Figure 2.1 shows the flow of data.

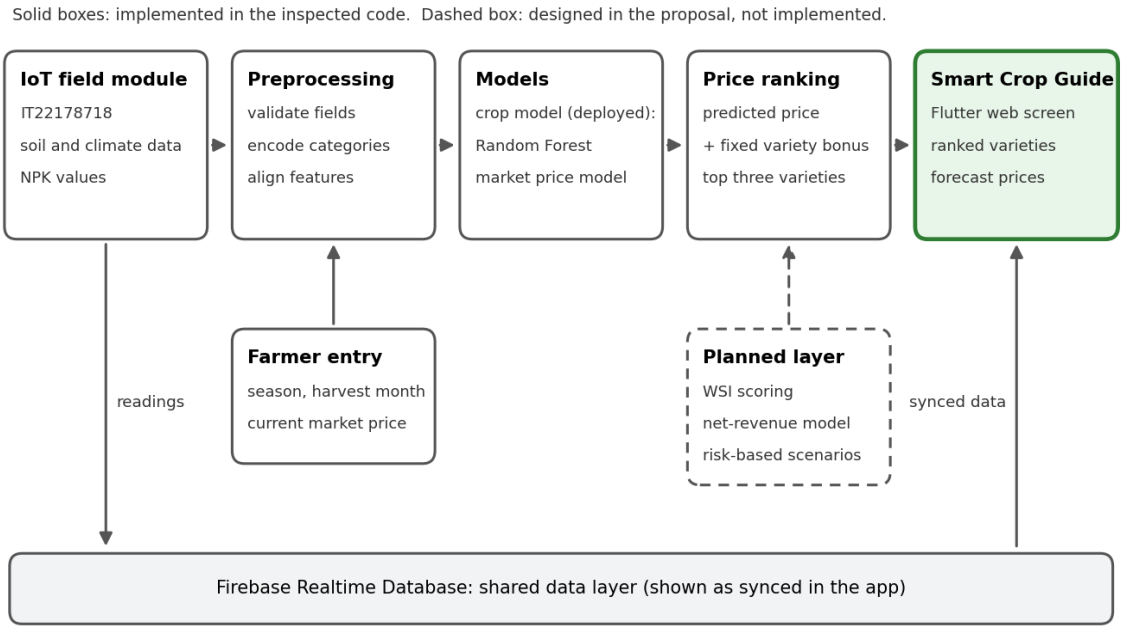


Figure 2.1: Component architecture: inputs to screen

Field readings and farmer entries pass through preprocessing and the models to the Smart Crop Guide screen. The planned layer adds suitability scoring, a validated price model and three-scenario ranking based on net revenue.

2.3 Data Inputs and Sources

The interface collects nine inputs. The training dataset, `paddy_dataset_final.csv`, has ten columns, as listed in Table 2.2.

Table 2.2: Input features and their roles

Input	Used in the Smart Crop Guide	Role
Soil moisture (%)	Yes	Field water condition
Temperature (°C)	Yes	Climate condition
Humidity (%)	Yes	Climate condition
Nitrogen, phosphorus, potassium (kg/ha)	Yes	Soil nutrient status
Season (Maha or Yala)	Yes	Cultivation season
Harvest month (1–12)	Yes (interface)	Timing of sale
Current paddy market price (Rs./kg)	Yes (interface)	Market signal
Growth stage	Dataset column only	Crop development stage
Variety (BG300, AT362)	Dataset column only	Candidate variety

Input	Used in the Smart Crop Guide	Role
Yield	Dataset column only	Outcome variable

The dataset follows the same schema as the group's shared paddy dataset: soil_moisture, temperature, humidity, nitrogen, phosphorus, potassium, growth_stage, season, variety and yield. In the visible rows, 13 of 25 yield values equal exactly 2800, which matches the repeated-value pattern reported for the shared dataset in the IT22266446 dissertation. The total record count is not visible in the supplied screenshot. The IT22266446 dissertation reports 12,437 records for a dataset with this schema. The origin of the repeated 2800 value has not been established.

Two observations shape the design. First, the dataset has no price or harvest-month column, so market forecasting cannot be learned from this file alone. Second, the interface does not ask for growth stage, although the dataset contains it. Both points are treated as limitations in Chapter 3.

The proposal planned climate data from the Sri Lanka Meteorological Department, labels from Department of Agriculture and Department of Census and Statistics records, and price history from the Hector Kobbekaduwa Agrarian Research Institute. The supplied evidence does not show these feeds being integrated, so they are reported as planned, not implemented.

2.4 Data Preprocessing and Feature Preparation

The backend stores the expected input columns of each model in two feature lists, `crop_features.pkl` and `market_features.pkl`, and uses them to build input tables in the order the models expect. Categorical inputs such as season are encoded to match the stored feature lists. The service reads the request fields and constructs the model input from them. The split ratio, random seed and missing-value handling used in training are not reproduced in the evidence supplied for this dissertation and are listed as a limitation.

2.5 Recommendation Models

Three classifiers were trained on the same features and compared on the same test set.

- **Random Forest** averages many randomized trees and resists overfitting on noisy tabular data [6].
- **XGBoost** adds trees sequentially to correct earlier errors, with regularization [7].
- **Decision Tree** is a single interpretable tree used as a simple baseline [8].

The common formulation is given in Equation 2.1.

$$\hat{c} = f_M(X_{soil},\ X_{climate},\ X_{season}) \qquad (2.1)$$

where ĉ is the recommended class, f is the trained model M, and the inputs are soil nutrients and moisture, temperature and humidity, and the season. The primary metric is accuracy. The backend loads the Random Forest as its crop model (`random_forest_model.pkl`). The supplied evidence does not state the exact definition of the classification target.

2.6 Market Forecast and Scenario Design

Implemented first version. For each candidate variety, the backend asks the market model (`market_prediction_model.pkl`) for a predicted price, adds a fixed variety bonus (BG300 +8, AT362 +3, BG352 +0) and sorts the candidates by predicted price from highest to lowest. The interface shows the top three results as a High Profit recommendation, a Balanced rank #2 and a Safe rank #3. The labels therefore follow price rank only.

Planned design. The proposal defines a Weighted Suitability Index (WSI) that scores each candidate variety against Department of Agriculture optimal ranges, as in Equation 2.2, and a projected net revenue per acre, as in Equation 2.3.

$$WSI_v = \Sigma\, w_i \cdot s_i(x_i), \;\; \text{summed over i} = 1 \text{ to n} \qquad (2.2)$$

where s_i scores how close field value x_i is to the optimal range for variety v and w_i is its weight. Varieties that fail minimum thresholds are removed.

$$NR_v = P_{forecast} \times Y_v - C_{input} \qquad (2.3)$$

where P is the forecast price per kg, Y is the expected yield and C is the input cost. Table 2.3 contrasts the intended scenario logic with the current implementation.

Table 2.3: Scenario logic: planned design and current implementation

Rank	Scenario	Planned logic	Current implementation
1	High Profit	Highest net revenue that passes the agronomic threshold	Highest predicted price after the fixed bonus
2	Balanced	Best combination of WSI and moderate profit	Second highest predicted price
3	Safe	Highest WSI and lowest field risk	Third highest predicted price

The WSI and net-revenue calculations do not appear in the inspected code. They remain design intentions.

2.7 Software Architecture and Implementation

The component has three parts. The client is AgriSmart Paddy, a Flutter application run as a web app in Microsoft Edge in debug mode. Its sidebar offers Farm Overview, Yield Doctor, Smart Crop Guide, Soil & Fertilizer and Leaf Health Scanner. The Smart Crop Guide screen, titled "Paddy Variety & Market Guide", has four input groups (soil and climate; NPK

nutrients; season and harvest timing; current market price) and a button labelled "Find Best Crops & Market Forecast". Results appear in a "Best Crop Recommendations" panel. The Flutter project contains the files `crop_recommendation_screen.dart`, `main_navigation_screen.dart` and `dashboard_screen.dart`.

The server is a Flask Blueprint named `crop_recommendation` in `backend-python/crop_reccomend/app.py`. It loads the crop model, the market model and the two feature lists with joblib, exposes a status route that returns "Paddy Crop Recommendation API Running", and provides `POST /predict`. A Firebase Realtime Database connection is shown as synced in the sidebar.

The proposal planned React.js, FastAPI, PostgreSQL, TensorFlow, AWS IoT Core and a Hostinger VPS. The implementation instead uses Flutter, Flask, joblib-stored tree-based models and Firebase, in line with the rest of the group's application. This change is also noted in Section 3.5.

2.8 Functional and Non-Functional Requirements

Table 2.4: Functional requirements

ID	Requirement
FR1	Accept soil, climate, nutrient, season, harvest-month and price inputs
FR2	Obtain field values from the IoT module through the shared platform
FR3	Predict a suitable variety using the trained classifier
FR4	Forecast market price for each candidate variety
FR5	Rank the top three varieties and label them as scenarios
FR6	Display recommendations and forecast prices in the interface
FR7	Update results when new sensor or price data arrive
FR8	Provide the current market price to the harvest-yield module (planned)

Table 2.5: Non-functional requirements

Area	Requirement
Performance	Return a recommendation within 3 seconds
Usability	Simple screens, clear results, Sinhala and English support
Reliability	Consistent output for equal inputs; tolerate missing sensor data
Security	Protect farmer and sensor data; restrict administration
Scalability	Support many farms and later datasets or modules
Maintainability	Allow models and price data to be updated without changing other modules

Listing a requirement here does not mean it has been tested.

2.9 Testing Strategy

Model testing reports accuracy, precision, recall, F1-score and a confusion matrix for the classifier, and R^2, MAE and RMSE for the price model, using held-out data. Functional testing checks valid input, missing and out-of-range values, button behaviour, result display and Firebase synchronization. Integration testing checks that the screen reaches the service and that every interface input is used by the backend. Usability testing is planned with 10 farmers. Measured results must come from actual runs.

2.10 Commercialization Aspects

Primary users are small and medium paddy farmers and cooperatives. Secondary users include extension officers, input suppliers, agri-finance institutions and buyers. The proposal outlines a free tier (soil suitability check and top two suggestions from historical averages) and a paid tier (price and demand forecasting and top five ranked recommendations). Pricing and demand are not yet evidenced, and the component should not be presented as a validated commercial product.

2.11 Methodological Limitations

The findings depend on a dataset whose provenance is incomplete, a small evaluation set, a classification target that is not documented in the supplied evidence, and a price model with no reported training data or metrics. The ranking uses fixed variety bonuses, and the suitability and net-revenue scoring are designed but not implemented.

2.12 Chapter Summary

The chapter set out the DSR approach, the inputs and dataset, the three compared models, the price-based ranking and the planned scoring, the Flutter, Flask and Firebase implementation, the requirements and the testing strategy. Chapter 3 reports what the evidence shows.

3. RESULTS AND DISCUSSION

The evidence for this chapter consists of a dataset screenshot, a model-accuracy chart, two screenshots of the Smart Crop Guide screen (before and after a recommendation), two Flutter console captures and three code screenshots from the Flask backend. These show that the interface and backend run, that three classifiers were trained and compared, and that a market model produces ranked prices. They do not include metrics for the market model, measured response times or a farmer usability study, so those items are reported as pending and no proposal target is presented as achieved.

3.1 Implemented Artifact Results

The Smart Crop Guide screen runs as a Flutter web application on localhost. The console capture in Figure 3.1 shows the device list (Windows desktop, Chrome 149 and Edge 155), the selection of Edge, and `lib\main.dart` launching in debug mode. Figure 3.2 shows the app linked to the debug service, the Dart VM service and DevTools available, and the application starting from `web_entrypoint.dart`.

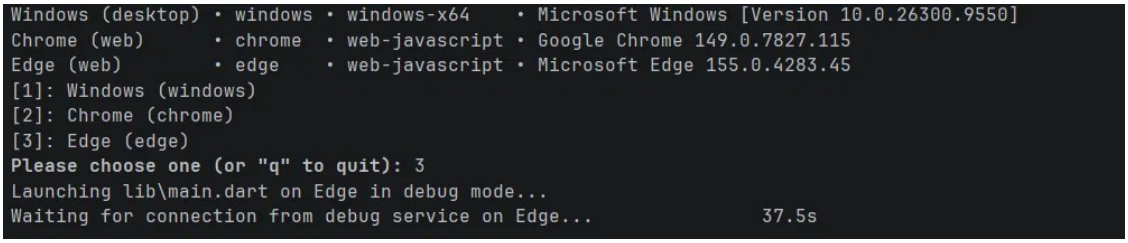


Figure 3.1: Flutter launch on Edge in debug mode (device selection)



Figure 3.2: Flutter debug service and application start

The console also reports that no Noto font set was found for all missing characters. This does not stop the app, but it matters for the requirement of Sinhala and English support: Sinhala text may render as empty boxes until a font asset is added.

Figure 3.3 shows the Smart Crop Guide screen before a recommendation is requested. The form holds moisture 65%, temperature 28 °C, humidity 80%, nitrogen 90, phosphorus 40, potassium 40 kg/ha, season Yala, harvest month 4 and a price of 120. The results panel reads "No recommendation generated". Three small interface issues are visible: the price field shows a dollar sign while its label says Rs./kg, the helper text says "Click Recommend" while the button reads "Find Best Crops & Market Forecast", and the example pairs Yala with harvest month 4, which should be checked against normal Yala harvest timing.

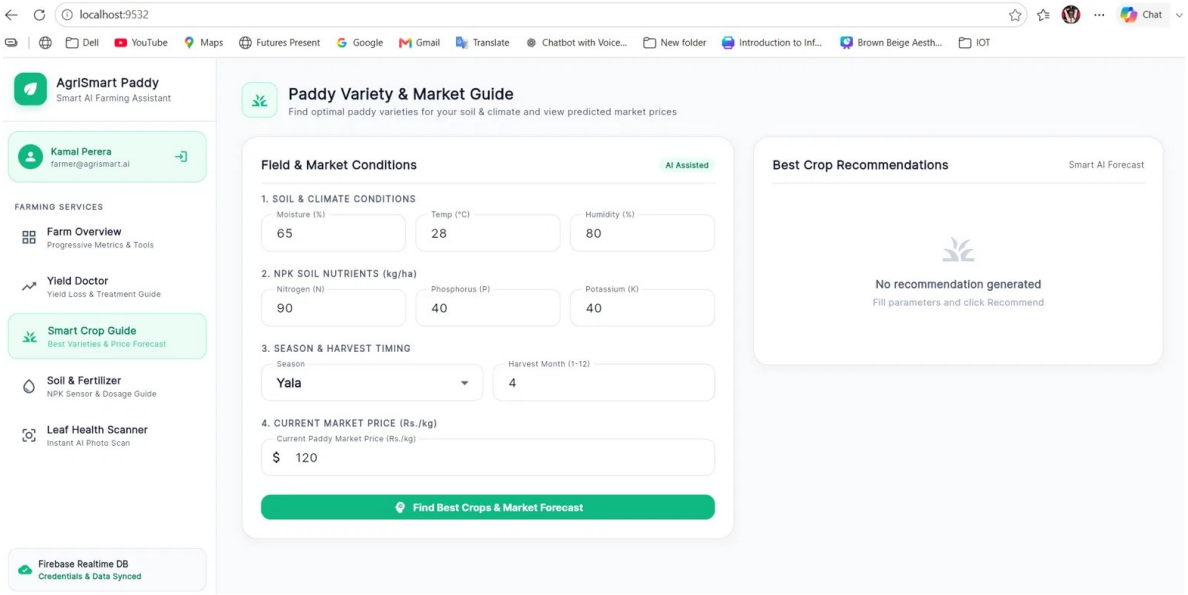


Figure 3.3: Smart Crop Guide screen before a recommendation is generated

Figure 3.4 shows the screen after the button is pressed with moisture 40%, temperature 34 °C, humidity 80%, nitrogen 9 (shown as 09), phosphorus 34, potassium 40 kg/ha, Yala season, harvest month 7 and a price of 120. The interface displays three ranked varieties: BG300 as the High Profit recommendation at Rs. 126.87 per kg, BG352 as Balanced rank #2 at Rs. 118.87 per kg, and AT362 as Safe rank #3 at Rs. 117.53 per kg. This shows that the request-to-result flow works for this input. It does not show that the recommendation is correct.

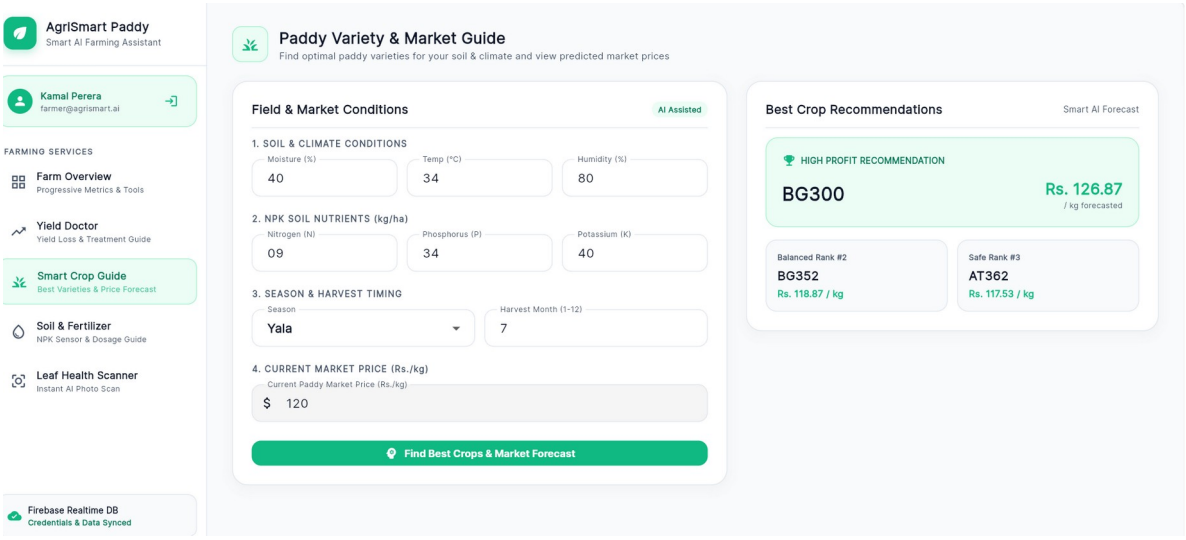
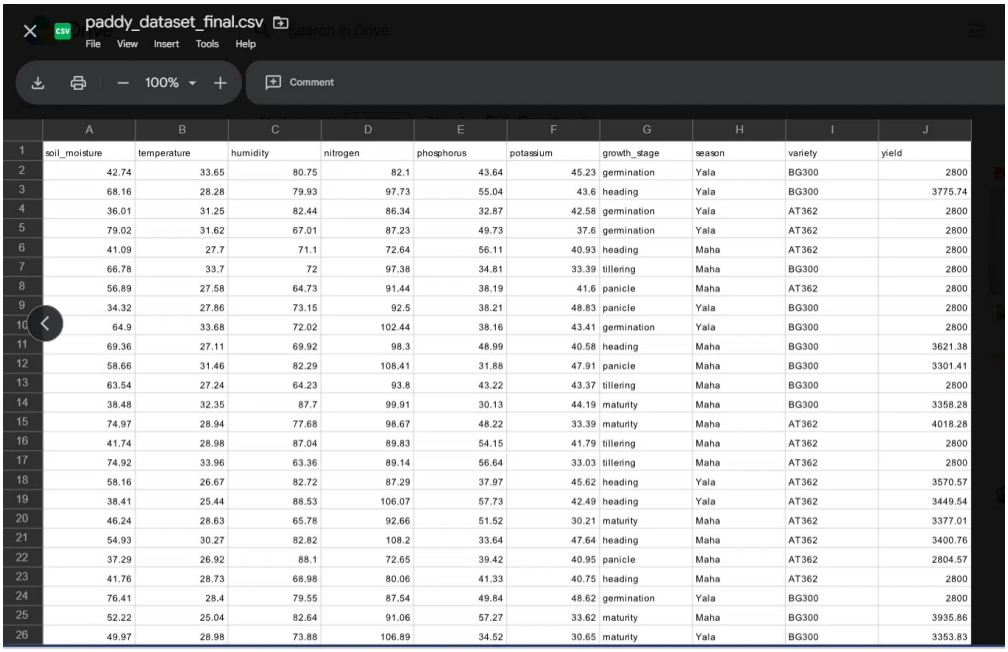


Figure 3.4: Smart Crop Guide screen showing the generated variety recommendation

3.2 Dataset Analysis

Figure 3.5 shows `paddy_dataset_final.csv` with ten columns: soil_moisture, temperature, humidity, nitrogen, phosphorus, potassium, growth_stage, season, variety and yield. Only 25 data rows are visible, so the figures below describe that sample and not the full file.



	A	B	C	D	E	F	G	H	I	J
1	soil_moisture	temperature	humidity	nitrogen	phosphorus	potassium	growth_stage	season	variety	yield
2	42.74	33.65	80.75	82.1	43.64	45.23	germination	Yala	BG300	2800
3	68.16	28.28	79.93	97.73	55.04	43.6	heading	Yala	BG300	3775.74
4	36.01	31.25	82.44	86.34	32.87	42.58	germination	Yala	AT362	2800
5	79.02	31.62	67.01	87.23	49.73	37.6	germination	Yala	AT362	2800
6	41.09	27.7	71.1	72.64	56.11	40.93	heading	Maha	AT362	2800
7	66.78	33.7	72	97.38	34.81	33.39	tillering	Maha	BG300	2800
8	56.89	27.58	64.73	91.44	38.19	41.6	panicle	Maha	AT362	2800
9	34.32	27.86	73.15	92.5	38.21	48.83	panicle	Yala	BG300	2800
10	64.9	33.68	72.02	102.44	38.16	43.41	germination	Yala	BG300	2800
11	69.36	27.11	69.92	98.3	48.99	40.58	heading	Maha	BG300	3621.38
12	58.66	31.46	82.29	108.41	31.88	47.91	panicle	Maha	BG300	3301.41
13	63.54	27.24	64.23	93.8	43.22	43.37	tillering	Maha	BG300	2800
14	38.48	32.35	87.7	99.91	30.13	44.19	maturity	Maha	BG300	3358.28
15	74.97	28.94	77.68	98.67	48.22	33.39	maturity	Maha	AT362	4018.28
16	41.74	28.98	87.04	89.83	54.15	41.79	tillering	Maha	AT362	2800
17	74.92	33.96	63.36	89.14	56.64	33.03	tillering	Maha	AT362	2800
18	58.16	26.67	82.72	87.29	37.97	45.62	heading	Yala	AT362	3570.57
19	38.41	25.44	88.53	106.07	57.73	42.49	heading	Yala	AT362	3449.54
20	46.24	28.63	65.78	92.66	51.52	30.21	maturity	Maha	AT362	3377.01
21	54.93	30.27	82.82	108.2	33.64	47.64	heading	Maha	AT362	3400.76
22	37.29	26.92	88.1	72.65	39.42	40.95	panicle	Maha	AT362	2804.57
23	41.76	28.73	68.98	80.06	41.33	40.75	heading	Maha	AT362	2800
24	76.41	28.4	79.55	87.54	49.84	48.62	germination	Yala	BG300	2800
25	52.22	25.04	82.64	91.06	57.27	33.62	maturity	Maha	BG300	3935.86
26	49.97	28.98	73.88	106.89	34.52	30.65	maturity	Yala	BG300	3353.83

Figure 3.5: Paddy dataset (first 25 records)

In the visible rows, soil moisture ranges from 34.3 to 79.0, temperature from 25.0 to 33.96 °C, humidity from 63.4 to 88.5, nitrogen from 72.6 to 108.4, phosphorus from 30.1 to 57.7 and potassium from 30.2 to 48.8. Growth stages are germination, tillering, panicle, heading and maturity. Seasons are Maha and Yala, and varieties are BG300 and AT362. The nitrogen value of 9 used in Figure 3.4 is far below the visible range and is probably a typing error for 90.

The target needs attention. In 13 of the 25 visible rows the yield equals exactly 2800, including rows at germination, tillering and panicle stages, and no visible value is below 2800. This pattern suggests a lower bound applied to the target, not natural yield variation. The IT22266446 dissertation reports the same repeated value for 6,300 of 12,437 records in a dataset with this schema. Any classification label derived from this target inherits that distortion. The file also contains no price or harvest-month column, and BG352, which the backend ranks, does not appear in the visible rows.

3.3 Model Comparison

Figure 3.6 and Table 3.1 compare the three classifiers on the same test set. The values are read from the bar heights of the chart and are approximate.

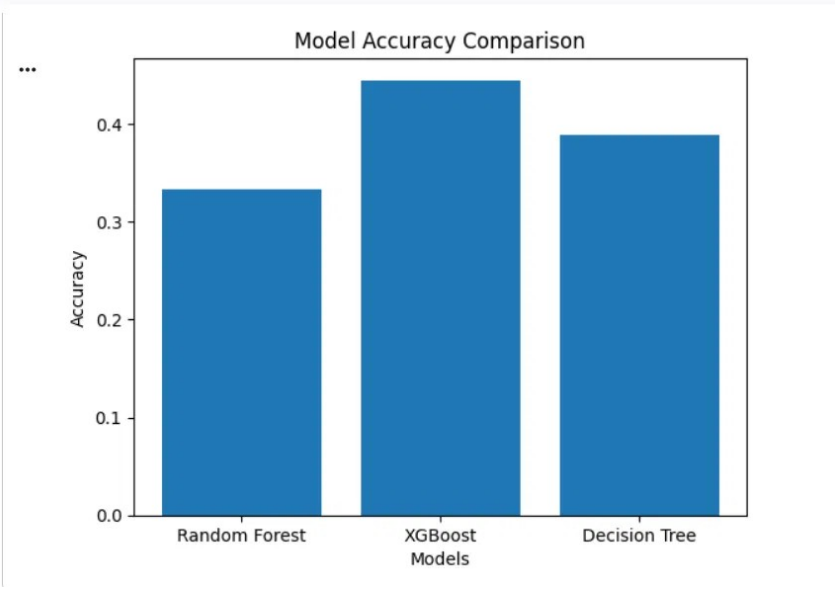


Figure 3.6: Model accuracy comparison (Random Forest, XGBoost, Decision Tree)

Table 3.1: Classifier accuracy (read from Figure 3.6)

Model	Accuracy (approx.)	Rank
XGBoost	0.44	1
Decision Tree	0.39	2
Random Forest	0.33	3

XGBoost is highest, but all three results are far below the 85% target in SO1. Five points explain how to read them.

1. **The test set appears very small.** The three values equal 8/18, 7/18 and 6/18, which suggests a test set of about 18 samples. One prediction then moves accuracy by about 5.6 points, and the gap between XGBoost and Random Forest is only two predictions. The ranking is not statistically reliable.
2. **Chance level matters.** With a balanced three-class target, chance accuracy is about 0.33, so Random Forest performs at chance. With a two-class target, 0.33 and 0.39 would be below chance. The target definition is not documented in the supplied evidence, so the numbers cannot be fully interpreted.
3. **The features carry signal for yield.** On the same schema, the IT22266446 notebook reports a Random Forest yield-regression R^2 of 0.9161 on 2,488 test records. The weak classification accuracy therefore points to the label design and test size, not to useless features. This is a hypothesis to test, not a conclusion.

4. **The ranking differs between tasks.** Random Forest was best for yield regression in that notebook but last here. This is consistent with high variance on a tiny test set.
5. **The deployed model is not the best one.** The backend loads `random_forest_model.pkl` as its crop model, although XGBoost scored highest in this comparison. Because the difference is within two predictions, either choice can be defended, but the reason for using Random Forest should be documented.

Only accuracy is available. Precision, recall, F1-score, a confusion matrix and cross-validation scores were not supplied and are not inferred.

3.4 Market Forecast and Scenario Ranking

The backend loads a market model, `market_prediction_model.pkl`, with its feature list. For each candidate variety it predicts a price, adds a fixed bonus, rounds the value to two decimals and sorts the results from highest to lowest (Appendix D). The output in Figure 3.4 follows from this logic. BG300 and BG352 have the same base predicted price of Rs. 118.87. BG300 ranks first only because of its bonus of 8. AT362 has a base price of Rs. 114.53, which becomes 117.53 with its bonus of 3. The labels "Balanced" and "Safe" therefore describe positions in a price ranking, not agronomic risk.

The bonus values are implementation assumptions. They are not learned or validated quantities and they decide the winner in the example. The supplied evidence also does not contain the price training data or any price-model metrics (R^2, MAE, RMSE), and the inspected code does not compute the WSI, the net revenue or revenue projections. SO2 is therefore implemented but unevaluated, SO3 is implemented in a first price-based form, and the proposal targets of $R^2 \geq 0.80$ and ranking consistency ≥ 90% remain unmeasured.

3.5 Integration with the Wider System

The Flask service exposes `POST /predict`, and the Flutter project contains the matching screen and navigation files. The sidebar shows a Firebase Realtime Database connection reported as synced, and the app hosts the other modules (Farm Overview, Yield Doctor, Soil & Fertilizer, Leaf Health Scanner). This supports the claim that the screen sits inside the shared platform. The architecture differs from the proposal, which planned React.js, FastAPI, PostgreSQL, TensorFlow and AWS IoT Core.

The inspected request-parsing code reads seven fields: soil moisture, temperature, humidity, nitrogen, phosphorus, potassium and season (Appendix C). The harvest month and the current market price that the screen collects are not read in that segment. Either they are used elsewhere in the code or they are ignored, so the interface may collect inputs that do not influence the result. This must be checked before the integration is described as complete.

End-to-end flow from the IoT module to a recommendation has not been demonstrated. The planned link that provides the market price to the harvest-yield module is not yet implemented, because the IT22266446 dissertation does not use market price in its decision route.

3.6 Functional Test Design

Table 3.2: Functional and integration test cases

ID	Scenario	Expected result	Evidence status
TC01	Complete valid input and button press	Recommendation and forecast displayed	Partly verified (Figure 3.4)
TC02	Empty or missing field	Clear validation message	Runtime evidence required
TC03	Out-of-range value (for example nitrogen 9 or humidity above 100)	Input rejected or flagged	Not yet; nitrogen 9 was accepted
TC04	Season and harvest month mismatch	Warning shown	Runtime evidence required
TC05	Screen load	Input form and empty-state panel shown	Verified (Figure 3.3)
TC06	Firebase telemetry load	NPK values populated when data exist	Runtime evidence required
TC07	Repeated identical requests	Identical ranking each time	Runtime evidence required
TC08	Harvest month and price change	Result changes with these inputs	Runtime evidence required

3.7 Non-Functional Evaluation

Table 3.3: Evidence-based evaluation summary

Item	Target	Actual result	Interpretation
Classification accuracy	≥ 85%	About 0.33–0.44	Target not met
Market price R^2	≥ 0.80	Not reported	Evaluation required
Response time	≤ 3 s	Not measured	Measurement required
Ranking consistency	≥ 90%	Not measured	Measurement required
Ranking basis	Suitability and net revenue	Predicted price plus fixed bonus	Partly met
Usability	≥ 80% task completion, 10 farmers	Not conducted	User study required
Sinhala and English support	Both languages shown	Font warning in console	Fix and verify

3.8 Research Findings

1. The Smart Crop Guide interface and Flask service exist and produce a ranked output for a test input, inside the shared Flutter application with Firebase connected.
2. Three tree-based classifiers were compared. XGBoost scored highest at about 0.44, but none approaches the 85% target, and the evaluation set appears too small to rank them reliably.
3. The deployed crop model is Random Forest, the lowest-scoring of the three.
4. The dataset target contains a large block of repeated values at 2800, and the dataset has no price or month data.
5. A market model produces predicted prices, but the ranking depends on fixed variety bonuses and no price-model metrics are available.
6. The WSI, net-revenue scoring and risk-based scenarios are not implemented, and the harvest month and price inputs may not reach the backend.
7. Performance and usability claims cannot yet be made.

3.9 Comparison with Existing Approaches

Compared with static advisory tools, the component accepts field values and a market price on one screen and returns ranked prices. Compared with the Nigerian Random Forest study [5], it uses Sri Lankan varieties and seasons but achieves far lower accuracy, which shows that a localized dataset alone does not guarantee performance. Compared with a full economic optimizer, it is incomplete because net revenue, cost and risk logic are not implemented.

3.10 Discussion of Research Objectives

Table 3.4: Objective achievement matrix

Objective	Evidence	Status	Remaining work
SO1 Suitability classifier	Three models trained; accuracy 0.33–0.44	Implemented, target not met	Define the label, enlarge the test set, report full metrics
SO2 Market price model	Market model artefact and predicted prices (Figure 3.4)	Implemented, no metrics	Document training data, report R^2, MAE and RMSE
SO3 Three-scenario ranking	Top three varieties ranked by price plus fixed bonus	First version implemented	Replace bonuses with WSI and net revenue; verify consistency
SO4 Application integration	Flutter screen, Flask route, Firebase sync	Implemented, partly verified	Confirm all inputs reach the backend; measure latency; fix interface issues
SO5 Evaluation	Accuracy chart, screenshots, code evidence	Partial	Add metrics, tests and usability study

3.11 Limitations

The main limitations are the very low classification accuracy, the small evaluation set, the repeated target value, the undocumented classification target, the missing price-model data and metrics, the fixed variety bonuses, the unimplemented suitability and net-revenue scoring, the possible mismatch between interface inputs and backend inputs, the missing Sinhala font, and the absence of latency and usability evidence.

3.12 Practical Implications

The screen demonstrates how a farmer could enter field conditions and a price and receive ranked variety guidance in one place. Until accuracy, the price model and the ranking logic are validated, the output should be treated as a prototype demonstration and not as advice for real planting decisions.

3.13 Commercialization Findings

The free-and-paid tier concept is feasible in principle because the component is modular. Commercial claims require validated accuracy, price forecasts, user acceptance, cost modelling and data governance first.

3.14 Chapter Summary

The interface, backend and model comparison exist, and the market model produces ranked prices, but the classification accuracy is well below target and the ranking rests on fixed bonuses. The next chapter states the contribution, required corrections and future work.

4. CONCLUSION

This research set out to convert field conditions, season, harvest timing and market price into a paddy variety recommendation and market forecast for Sri Lankan farmers. The component delivers a working Smart Crop Guide interface, a Flask prediction service that returns three ranked varieties with forecast prices, and a first comparison of three tree-based classifiers. It does not yet deliver a recommendation accurate enough for farmer use, and its price model and ranking logic are not yet validated.

4.1 Summary of the Research

The literature showed that suitability, climate and market information are treated separately in existing tools. The research framed the gap as an integration problem and followed Design Science Research to build and evaluate a component for it. Random Forest, XGBoost and Decision Tree were compared on a paddy dataset, and XGBoost scored highest at about 0.44 accuracy. A market model supplies predicted prices that the backend adjusts with fixed variety bonuses and ranks.

4.2 Achievement of the Aim

The aim was to develop and evaluate a market-aware recommendation component. The development part is largely achieved at prototype level: the interface, the backend, the model comparison and the Firebase connection exist, and the system returns ranked output. The evaluation part shows that the accuracy target of 85% was not met and that price-model metrics, ranking consistency, latency and usability results are still missing.

4.3 Individual Contribution

The contribution of IT21811104 is the crop-recommendation and market layer of the R26-IT-066 DSS. It is distinct from the IoT monitoring (IT22178718), disease analysis (IT22261632) and harvest-yield (IT22266446) components. It is designed to receive field data from the IoT module and, in a planned extension, to supply the current market price to the yield module.

4.4 Key Technical Contribution

The main technical contribution is a farmer-facing screen and Flask service that bring soil, climate, season, harvest timing and price into one request and return ranked varieties with forecast prices, together with a documented baseline comparison of three classifiers. The baseline, including its weaknesses, gives later work a clear starting point.

4.5 Limitations and Required Corrections

Before the component can be relied on, the following corrections are needed.

1. Document the classification target and rebuild it so that it is not distorted by the repeated 2800 yield value.
2. Investigate the origin of the 2800 value and confirm the dataset size.
3. Enlarge the test set and report accuracy, precision, recall, F1-score, a confusion matrix and cross-validation scores. Explain or change the choice of Random Forest as the deployed model.
4. Document the price training data and report R^2, MAE and RMSE for the market model.
5. Replace the fixed variety bonuses with the WSI and net-revenue scoring, and verify ranking consistency and response time.
6. Make the backend use the harvest month and the current price, and add input-range and season-month validation.
7. Fix the interface issues: currency symbol, helper text and the Sinhala font asset.
8. Run end-to-end tests with the IoT module and carry out the farmer usability study.

These corrections do not invalidate the artefact. They define the work needed to move from a prototype to a validated decision-support component.

4.6 Future Work

Future work should collect larger field and price datasets covering more districts, varieties and seasons, review the price-forecasting literature, and model yield and price together so that the recommendation reflects expected revenue. It should add confidence intervals, explanations in Sinhala and Tamil, periodic model retraining and secure cloud deployment, connect the market price to the yield module, and extend the approach to other crops.

4.7 Final Conclusion

Market-aware variety recommendation is a sound idea, and the working prototype shows how it can be presented to farmers. The present evidence supports the interface, the backend flow and a baseline experiment, not a reliable recommendation. The dissertation therefore reports both what was built and the evidence-based limits of what the system can currently claim.

REFERENCES

[1] Department of Census and Statistics, Sri Lanka, "Paddy Statistics – Maha 2023/2024 and Yala 2024," Ministry of Agriculture, Colombo, 2024. [Online]. Available: https://www.statistics.gov.lk/Agriculture/StaticalInformation/PaddyStatistics

[2] O. Friha, M. A. Ferrag, L. Shu, L. Maglaras, and X. Wang, "Internet of Things for the future of smart agriculture: A comprehensive survey of emerging technologies," IEEE/CAA Journal of Automatica Sinica, vol. 8, no. 4, pp. 718–752, 2021, doi: 10.1109/JAS.2021.1003925.

[3] S. Wolfert, L. Ge, C. Verdouw, and M.-J. Bogaardt, "Big data in smart farming – A review," Agricultural Systems, vol. 153, pp. 69–80, 2017, doi: 10.1016/j.agsy.2017.01.023.

[4] K. G. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine learning in agriculture: A review," Sensors, vol. 18, no. 8, Art. no. 2674, 2018, doi: 10.3390/s18082674.

[5] I. Abdurrahman and A. M. Miyim, "Machine Learning Model for Predicting Rice Crop Yield: A Case Study in Hadejia and Auyo, Nigeria," UMYU Scientifica, vol. 4, no. 1, pp. 239–249, 2025.

[6] L. Breiman, "Random forests," Machine Learning, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[8] L. Breiman, J. Friedman, R. A. Olshen, and C. J. Stone, Classification and Regression Trees. Belmont, CA: Wadsworth, 1984.

[9] Y. Wang, U. S. R. Dhamodharan, N. Sarwar, F. A. Almalki, and D. Mohan, "A hybrid approach for rice crop disease detection in agricultural IoT system," Discover Sustainability, vol. 5, Art. no. 99, 2024, doi: 10.1007/s43621-024-00280-5.

[10] A. Gumma et al., "Satellite-based assessment of crop growth stages in paddy-dominated Asian lowlands," International Journal of Remote Sensing, vol. 42, no. 5, pp. 1795–1816, 2021.

[11] D. G. Peng et al., "Modeling the impact of rice blast on yield under climate change scenarios in Asia," Global Food Security, vol. 28, 2021, Art. no. 100493.

[12] W. Chen et al., "Deep learning-based identification and severity assessment of rice blast disease," Computers and Electronics in Agriculture, 2024.

[13] N. Mandal et al., "Spectral characterization and severity assessment of rice blast disease using univariate and multivariate models," Frontiers in Plant Science, vol. 14, 2023, Art. no. 1067189, doi: 10.3389/fpls.2023.1067189.

[14] "eKeth: A Machine Learning-Based Mobile Platform to Facilitate the Paddy Cultivation Process in Sri Lanka," in Proc. 6th Int. Conf. Information Technology Research (ICITR), 2021, pp. 1–6, doi: 10.1109/ICITR54349.2021.9657468.

[15] G. Amarnath et al., "GeoGoviya: Smart Farming Platform for Climate-Smart Agriculture in Sri Lanka," International Water Management Institute, Colombo, Sri Lanka, 2022.

[16] Department of Agriculture, Sri Lanka, "Krushi Advisor: Agricultural Advisory Mobile Application," National Agriculture Information and Communication Centre, 2023. [Online]. Available: https://doa.gov.lk/

[17] Department of Agriculture, Sri Lanka, "Crop Recommendations and Fertilizer Guidelines," 2021.

[18] International Rice Research Institute, Standard Evaluation System for Rice, 5th ed. Los Baños, Philippines: IRRI, 2013.

[19] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," MIS Quarterly, vol. 28, no. 1, pp. 75–105, 2004, doi: 10.2307/25148625.

GLOSSARY

Term	Description
Maha, Yala	The two principal paddy cultivation seasons in Sri Lanka
BG 300, AT 362, BG 352	Paddy varieties referred to in this research
Weighted Suitability Index	Score of how well field conditions match a variety's optimal ranges (planned)
Net revenue	Forecast price times expected yield, minus input cost (planned)
Variety bonus	Fixed amount added to a variety's predicted price in the current backend
Decision-support system	Software that combines data, models and rules to assist human decisions
Baseline	A first reference result against which later improvements are compared

APPENDICES

APPENDIX A – EVIDENCE INDEX

Table A.1 lists the evidence supplied for this dissertation and what each item shows.

Table A.1: Evidence index

Figure	Evidence	Shows
3.1, 3.2	Flutter console captures	App launches on Edge in debug mode; Noto font warning
3.3	Smart Crop Guide screenshot (before)	Input form and empty-state results panel
3.4	Smart Crop Guide screenshot (after)	Three ranked varieties with forecast prices
3.5	paddy_dataset_final.csv	Ten-column dataset; repeated 2800 yield
3.6	Model accuracy chart	Random Forest, XGBoost and Decision Tree accuracy
B.1	Backend source code	Model loading and service setup
C.1	Backend source code	Request parsing and crop feature construction
D.1	Backend source code	Price prediction, variety bonus and ranking

APPENDIX B – SOURCE-CODE EVIDENCE: MODEL LOADING AND SERVICE SETUP

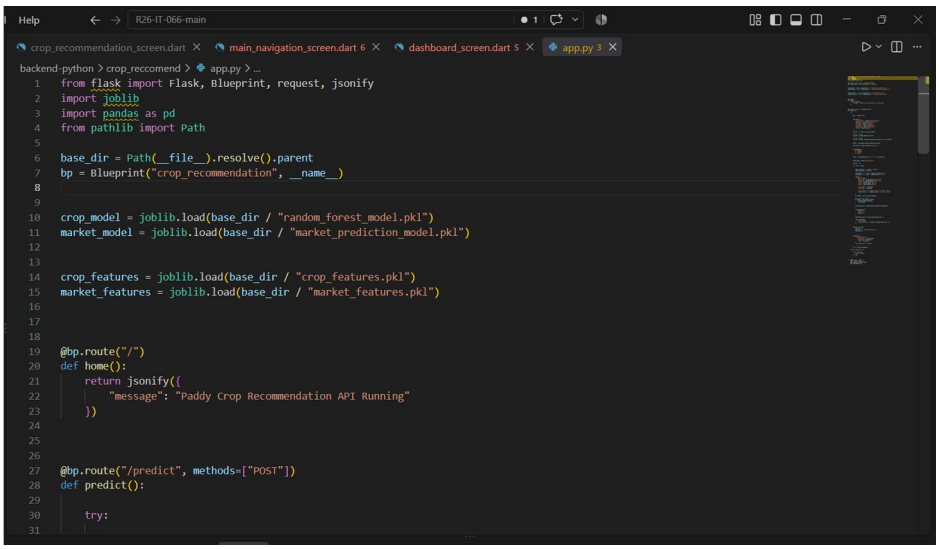


Figure B.1: Model loading and route definitions in the crop recommendation backend

The source code loads four artefacts: a crop model stored as `random_forest_model.pkl`, a market price model stored as `market_prediction_model.pkl`, and two feature-column lists, `crop_features.pkl` and `market_features.pkl`. The feature lists let the service build input tables in the column order each model expects. The file name shows that the deployed crop model is a Random Forest, although XGBoost scored highest in the

comparison in Section 3.3. The code also defines a status route and the `POST /predict` route.

APPENDIX C – SOURCE-CODE EVIDENCE: INPUT CONSTRUCTION



```
@bp.route("/predict", methods=["POST"])
def predict():

    try:

        data = request.json


        crop_input = {
            "soil_moisture": data["soil_moisture"],
            "temperature": data["temperature"],
            "humidity": data["humidity"],
            "nitrogen": data["nitrogen"],
            "phosphorus": data["phosphorus"],
            "potassium": data["potassium"],
            "season": data["season"]
        }
```

Figure C.1: Request parsing and crop feature construction

The endpoint reads the JSON body and builds the crop input from seven fields: soil moisture, temperature, humidity, nitrogen, phosphorus, potassium and season. In the visible segment, the harvest month and the current market price shown on the Smart Crop Guide screen are not read from the request. Whether these two inputs reach the market model elsewhere in the code, or are ignored, must be confirmed before the integration is described as complete.

APPENDIX D – SOURCE-CODE EVIDENCE: PRICE PREDICTION AND RANKING

```
        predicted_price = market_model.predict(df_sample)[0]


        variety_bonus = {
            "BG300": 8,
            "AT362": 3,
            "BG352": 0
        }

        predicted_price += variety_bonus.get(crop, 0)

        results.append({
            "crop": crop,
            "predicted_price": round(float(predicted_price), 2)
        })


    results = sorted(
        results,
        key=lambda x: x["predicted_price"],
        reverse=True
    )
```

Figure D.1: Price prediction, variety bonus and ranking logic

For each candidate, the market model returns a predicted price. The code then adds a fixed bonus of 8 for BG300, 3 for AT362 and 0 for BG352, appends the crop and its rounded price to the results, and sorts the results from highest to lowest price. The bonus values are implementation assumptions that have not been validated. The ranking in this segment is based on predicted price alone. It does not compute the weighted suitability index, the net revenue or the risk-based scenarios described in Section 2.6. BG352 also appears here although the dataset screenshot shows only BG300 and AT362.